

AsymMark: Weakly Asymmetric Behavioral Watermarking for LLM Agents

Abstract—Autonomous LLM agents act through tool calls, subgoal choices, and embodied decisions, so provenance must cover behavior trajectories rather than final text alone. Distribution-preserving behavioral watermarks are attractive for deployment because they embed provenance without distorting the agent’s policy. Existing verification models sit at two extremes: exact-channel verification assumes the verifier can replay the per-step behavior probabilities used at embedding, while selected-action-only verification has too little channel knowledge to recover a distribution-preserving payload. Realistic audits often fall between these extremes: verifiers may face model drift, closed or quantized endpoints, or APIs that expose only a truncated top- k view. Under this partial channel knowledge, exact-channel symmetric decoders such as AgentMark-F lose usable capacity because their bins depend on probability mass rather than on the rank information the verifier can reconstruct. We propose AsymMark, a weakly asymmetric framework that makes verifier knowledge an explicit design axis, and instantiate AsymMark-R using top- k rank order as the decoding invariant. AsymMark-R uses probabilities only at embedding time to preserve the agent policy; verification decodes from the selected behavior’s rank path and pools coded packets across audit windows. We report effective capacity to separate nominal embedding opportunity from capacity that survives the verifier’s view. On ALFWorld and ToolBench across DeepSeek and Gemini Flash, AsymMark-R preserves task utility and behavior-frequency stealth, retains substantially more capacity under rank-only verification, and recovers multi-bit payloads over pooled audit windows. Behavioral provenance should therefore be designed around the channel the verifier can actually reconstruct.

1. Introduction

Large language model (LLM) systems are rapidly shifting from passive text generators to agents that perceive context, plan, call tools, and act across multiple steps [1], [2], [3]. This shift changes what provenance must mean. For a chatbot, attributing the final string may be enough. For an agent, the consequential object is often the trajectory: which tool it selected, which subgoal it pursued, which API it invoked, or which embodied action it chose. These planning behaviors can affect safety, accountability, intellectual-property protection, and post-hoc auditing even when the final natural-language output is short or absent.

Content watermarking for LLM outputs has made substantial progress [4], [5], but it does not directly solve behavioral provenance. Planning behaviors are not token-

native: an LLM’s reasoning text may be compiled into structured tool calls or environment actions, discarding the lexical signal used by token watermarks. In response, AgentMark [6] formulated behavioral watermarking as distribution-preserving sampling over an explicit per-step behavior distribution. Its concrete construction, which we call AgentMark-F, embeds multi-bit identifiers while preserving the marginal behavior distribution, and it tolerates partial logs through erasure coding. That result establishes the utility-preserving behavior channel; the remaining question is what the verifier must know about that channel at audit time.

However, AgentMark-F is symmetric: decoding requires the verifier to reconstruct the same probability distribution used by the embedder. This is a strong assumption in real deployments. At the other extreme, a verifier that sees only selected actions lacks enough channel knowledge to decode a distribution-preserving payload. Many audits occupy the middle ground: the verifier may access a rounded top- k API, a model after provider-side updates, a different serving stack, or a proxy model. These views often preserve the relative order of plausible behaviors while shifting the numeric probabilities.

This is not only an implementation inconvenience; it is a structural verifier-channel failure mode. AgentMark-F’s differential bins are functions of probability masses. Two distributions can induce the same top- k order but move a cumulative boundary across a rank position, causing the same selected behavior to decode to a different bin. Thus a state-of-the-art exact-channel behavioral watermark can look strong under its native metric yet fail in the practically important rank-only audit channel. The lesson is broader than one implementation: behavioral watermarking should measure the channel available to the verifier, not only the capacity available to the embedder.

This paper asks whether behavioral provenance can exploit that middle ground. We formulate AsymMark, a weakly asymmetric behavioral watermarking framework for agent trajectories: the embedder may use the full behavior distribution P_t to preserve the agent policy, while the verifier decodes from an invariant reconstructed from its restricted view. This setting is weaker than exact-probability verification but stronger than selected-action-only verification. We use rank order as the first evaluated invariant for agent behavior channels because common audit interfaces expose candidate rankings, and rank is more stable than calibrated probability values while still more informative than the selected action alone.

We instantiate the framework as AsymMark-R, a top- k

rank-based behavioral watermark. The rank-splitting primitive is not claimed as a new steganographic primitive; it is adapted from weakly asymmetric rank-based steganography [7]. Our contribution is to make the weak-asymmetry abstraction explicit for AgentMark’s behavior channel, where candidate actions rather than text tokens carry provenance, and to specify the synchronization, coding, and audit protocol needed for logged agent trajectories. The encoder sorts behaviors by probability and recursively partitions the top- k list into even and odd ranks. At each level, a binary distribution-preserving primitive uses probability masses to choose a group without changing the marginal policy. The decoder, by contrast, never uses probability values: the selected behavior’s rank determines the path through the same interleaved partition tree, and odd branches reveal embedded bits. A fixed-budget deterministic random bit generator (DRBG) schedule ensures that encoder and decoder consume the same pseudorandomness even when no bit is embedded at a level.

We organize the evaluation around effective capacity,

$$C_{\text{eff}} = C_{\text{nom}} \times \text{DSR}, \quad (1)$$

where C_{nom} is the nominal number of recoverable coded bits exposed by the channel and DSR is the decode success rate under the verifier’s view. This yardstick separates a watermark that has high capacity only under exact probabilities from one that yields lower nominal capacity but survives a degraded verifier channel. It also makes packet scarcity explicit: individual traces contribute packets, and attribution is evaluated over pooled audit windows once enough independent equations accumulate.

The claim is intentionally scoped. The paper establishes a channel-knowledge tradeoff: top- k ranks can preserve decoder signal when exact probabilities are unavailable. It does not claim attribution from selected actions alone; audit-grade attribution still requires a pre-registered key/payload claim, calibrated rank reconstruction, and end-to-end RLNC recovery under the deployment logging stack.

Our contributions are:

- **Weakly asymmetric behavioral watermarking framework.** We formulate AsymMark as a framework that decouples the policy-preserving embedding channel from a verifier-reconstructible decoding invariant. The invariant must be recoverable from the verifier’s restricted view, independent of calibrated probability values, and sufficient to index the decoder’s partition path. This framing makes verifier-channel knowledge an explicit design object rather than an implicit replay assumption.
- **Rank instantiation and audit protocol.** We instantiate AsymMark as AsymMark-R, using top- k rank order as the decoding invariant. The rank primitive is adapted from weakly asymmetric steganography [7]; the new contribution is its agent-behavior instantiation with fixed $\lceil \log_2 n_t \rceil$ DRBG synchronization and deterministic RLNC over logged trajectories.
- **Verifier-channel failure mode, formalized and measured.** We identify probability-bin instability as a distinct

robustness dimension, formalize it in Lemma 1, and empirically quantify it: under rank-stable top- k truncation, AgentMark-F’s bins still change on 13.7% of ALFWorld DeepSeek steps at top-10 and 56.8% at top-4, with matching decoded-bit flips of 11.1% and 51.3%.

- **Effective-capacity and audit-window evidence model.** We adopt $C_{\text{eff}} = C_{\text{nom}} \times \text{DSR}$ to score capacity that survives the verifier’s channel, and model payload evidence as cumulative packet recovery over pooled audit windows through the packet-scarcity bound in Proposition 3.
- **Verifier-channel evaluation on agent trajectories.** We evaluate on ALFWorld [8] and ToolBench [9] across DeepSeek and Gemini Flash, showing that AsymMark-R preserves utility and behavior-frequency stealth while retaining usable capacity when the verifier degrades from exact probabilities to top- k ranks. Supplemental calibration studies cover rank noise, semantic rewrites, and live cross-model reranking.

The rest of the paper proceeds as follows. Section 2 motivates behavioral watermarking and weak asymmetry. Section 3 defines the agent behavior channel, adversary, and verifier knowledge. Section 4 presents the rank encoder, decoder, synchronization, and coding layer. Sections 5 and 6 analyze security and evaluate utility, stealth, and rank-only recovery. Section 7 discusses operational limits, Section 8 covers related work, Section 9 covers validity and ethics, and Section 10 concludes. The appendices provide the optional depth behind the main claims: distribution preservation, DRBG synchronization, RLNC recovery, rank stability, and artifact mapping.

2. Background and Motivation

2.1. Behavioral Watermarking for Agents

An LLM agent operates over a trajectory of observations, planning behaviors, and execution actions. At each step t , the agent chooses a behavior b_t from a finite admissible set $\mathcal{B}_t$. Examples include ALFWorld navigation actions, ToolBench API calls, or social-simulation behaviors in OASIS [10]. AgentMark models this choice as sampling from an elicited behavior distribution P_t and replaces ordinary sampling with a distribution-preserving encoder:

$$\hat{b}_t \leftarrow \text{Enc}(P_t, r_t), \quad \Pr[\hat{b}_t = b] = P_t(b). \quad (2)$$

The distribution-preservation requirement is central. Agent trajectories are long-horizon and stateful; even small per-step distribution shifts can compound into task drift, failures, or visibly abnormal behavior.

AgentMark’s verifier observes the selected behavior together with the same elicited distribution P_t for each retained step. Its robustness experiments therefore address missing records: erasure, truncation, and surface rewrites that may remove or perturb parts of the log. They do not remove the exact-channel assumption for the steps that remain. Our work starts from this interface and changes a different axis: the verifier may retain the trajectory and

selected behaviors but lose calibrated access to the original per-step probabilities.

2.2. The Symmetry Bottleneck

The concrete AgentMark-F construction uses differential recombination over P_t followed by cyclic-shift encoding. This is efficient when both embedder and verifier share exact probabilities. The same property makes it brittle under probability perturbation: if the verifier’s reconstructed P'_t differs from P_t , recombined bins may differ, so the selected behavior is interpreted under the wrong bin structure.

The weakness is not merely numerical. Many verification interfaces intentionally withhold calibrated probabilities. Auditors may receive ranked candidates, top- k choices, or sampled logs. Even when probabilities are available, temperature scaling, quantization, floating-point differences, model upgrades, and fine-tuning can perturb values. These transformations frequently preserve a useful invariant: the order of high-probability behaviors.

This observation is especially natural for agents. Candidate behaviors are usually semantic alternatives produced by a planning prompt: a tool call, a navigation operation, a search/refine step, or a social action. A downstream auditor may be able to ask a model to rank plausible next behaviors from the same logged context even if the exact sampling distribution is hidden. In this sense, behavior-level verification has a useful middle ground that token-level watermarking often lacks: ranks are easier to reproduce than calibrated probabilities, and more informative than the selected action alone.

This gives a three-level verifier-knowledge hierarchy. Exact-channel behavioral watermarking assumes calibrated probability replay and gives high capacity when the assumption holds. Selected-action-only verification minimizes verifier knowledge but cannot decode a distribution-preserving payload in our setting. AsymMark targets the useful middle: it keeps AgentMark’s behavior-trajectory carrier and preserves the behavior policy P_t , but lets the decoder depend on a verifier-reconstructible invariant rather than exact probability replay. AsymMark-R is the rank-order instance of this framework, using top- k rank σ_t as the verifier channel rather than claiming the rank primitive itself is new.

2.3. Weak Asymmetry

Provably secure steganography traditionally separates symmetric settings, where sender and receiver share channel distributions, from asymmetric settings, where the receiver has no channel oracle [11], [12]. Weak asymmetry occupies the middle ground: the sender has the full channel, while the receiver has partial but nontrivial channel knowledge [7]. We lift this narrative from text steganography to behavioral watermarking. The embedder uses P_t to preserve the agent policy; the verifier uses a lower-dimensional invariant that is easier to reconstruct than calibrated probabilities. In AsymMark-R, that invariant is $\sigma_t = \text{argsort}_k(P_t)$.

3. Problem Formulation

3.1. Agent Behavior Channel

For a trajectory of length T , each step t has a behavior set $\mathcal{B}_t = \{b_{t,1}, \dots, b_{t,N_t}\}$ and a behavior distribution P_t induced by the agent and environment context. A watermarking key K_{sh} and step context h_t derive per-step pseudo-randomness. The embedder encodes a payload $m \in \{0, 1\}^L$ into selected behaviors $\hat{b}_1, \dots, \hat{b}_T$ while preserving P_t at each step.

The verifier observes a possibly incomplete set of indices $I \subseteq \{1, \dots, T\}$ and, for each observed step, the selected behavior $\hat{b}_t$ plus channel side information. In the symmetric setting, the side information is P_t or an exact reconstruction. In the weakly asymmetric setting, the side information is a verifier-reconstructible invariant; in AsymMark-R, it is the top- k ordering σ_t .

Verifier knowledge. We distinguish three verifier views. In the *exact view*, the verifier receives P_t and can run the same decoder as the embedder. In the *rank view*, the verifier receives only $\sigma_t^k = (b_{(1)}, \dots, b_{(k)})$, the ordered list of the top- k behaviors. In the *selected-only view*, the verifier receives only $\hat{b}_t$ and no usable channel side information. The exact view is a useful upper bound; the rank view is the deployment target of AsymMark-R; the selected-only view is a negative control showing that behavioral provenance cannot be recovered from actions alone without some channel side information.

For evaluation we instantiate these views as a channel-knowledge ladder. $L0$ is exact probability access. $L1$ – $L5$ progressively restrict the verifier to top-10, top-8, top-6, top-4, and top-2 ordered candidates. $L6$ is the selected-only negative control. This ladder is not a claim that every API exposes exactly these levels; it is a controlled way to ask how usable capacity changes as the verifier loses calibrated probability information.

Decodable invariants. AsymMark separates the embedding channel from the decoding channel through a per-step invariant $z_t = \phi(P_t, \mathcal{B}_t)$. A useful weak-asymmetric invariant must satisfy three conditions. First, it must be reconstructible from the verifier’s restricted view, such as a rounded top- k API or proxy scorer. Second, it must be independent of the numeric probability values used by the embedder to preserve $\Pr[\hat{b}_t = b] = P_t(b)$. Third, it must be sufficient to index the shared partition path that turns a selected behavior into coded bits. Top- k rank order satisfies these conditions for AsymMark-R: the verifier can reconstruct an ordered candidate list, probability-value perturbations do not change the path when the order is stable, and the selected behavior’s rank identifies the even/odd partition decisions. Other invariants are possible in principle; Section 7 returns to this broader design space.

Rank consistency. For two distributions P_t and P'_t over the same candidate list, define top- k rank consistency as

$$\text{argsort}_k(P_t) = \text{argsort}_k(P'_t). \quad (3)$$

This condition permits arbitrary probability-value changes as long as the verifier preserves the ordered top- k set. It covers monotone transformations such as temperature rescaling, but it also covers less structured API changes whose empirical effect is to leave ranks unchanged. When the equality fails, we measure degradation through rank-noise experiments rather than assuming correctness.

3.2. Adversary and Requirements

We consider three perturbations. First, *step erasure*: some logged steps are missing due to execution failures, platform deletion, or incomplete telemetry. Second, *distribution perturbation*: a verifier reconstructs P'_t whose numeric values differ from P_t . Third, *rank noise*: the verifier's top- k order disagrees with the embedder's order, modeling top- k instability or proxy-model disagreement.

The adversary observes watermarked trajectories and may know the algorithm and public parameters, including k , but not K_{sh} . This is the standard secret-key provenance setting: a platform or model owner embeds an identifier, while a later verifier checks a claimed key and payload. We do not rely on security through obscurity of the rank partition or coding scheme.

The adversary has two goals. A *removal* adversary tries to make a genuinely watermarked trajectory verify as inconclusive or negative by deleting steps, perturbing ranks, or changing behavior serialization. A *framing* adversary tries to make an unwatermarked or differently watermarked trajectory verify under another party's claim. Claim binding and false-positive thresholds address framing when the key is secret; arbitrary trajectory rewriting, compromised keys, and compromised logging infrastructure are outside the cryptographic claim and are handled as deployment assumptions.

The audit boundary is therefore explicit. The verifier needs a trajectory log whose step contexts and selected behaviors are bound by the embedding service or another trusted logging layer, plus a deterministic mapping from logged behaviors to verifier-side candidates. The watermark can tolerate missing steps and imperfect rank reconstruction, but it cannot distinguish a genuine erasure from a log rewritten by an untrusted recorder. We treat log integrity and candidate canonicalization as deployment prerequisites and evaluate the watermark channel conditional on those prerequisites.

A scheme should satisfy:

- **Utility preservation:** $\Pr[\hat{b}_t = b] = P_t(b)$ per step.
- **Weak-asymmetric correctness:** if the verifier reconstructs the same invariant z_t as the embedder, decoding returns the same bits; for AsymMark-R, this means $\text{argsort}_k(P'_t) = \text{argsort}_k(P_t)$.
- **Low false positives:** for a fixed pre-registered L -bit claim, a clean or unrelated trajectory should recover that exact payload with probability 2^{-L} when decoded bits are independent of the claim.
- **Erasures tolerance:** payload recovery should degrade gracefully when only a subset of steps is observed.

Detection statistic. For an audit window, the verifier decodes a multiset of coded bits and attempts to recover payload $\hat{m}$. A positive attribution requires $\hat{m} = m$ for a pre-registered claimed payload after packets have been accumulated over the window. A failed recovery is treated as negative only when the verifier has enough rank-consistent evidence and independent equations; otherwise it is inconclusive. Payload recovery is therefore the provenance claim: before the pooled window has enough independent equations for full RLNC recovery, the audit should keep collecting evidence or report an inconclusive outcome.

3.3. Non-Goals

AsymMark-R is not a general-purpose watermark removal defense. If an adversary controls the agent and can arbitrarily rewrite the trajectory, no behavioral watermark can force attribution without an external logging root of trust. Our target is post-hoc verification for trajectories that are generated by an honest embedder but later audited through an imperfect channel interface. We also do not assume the verifier can re-run the exact original model. The verifier may use a same-family model, a refreshed endpoint, or a restricted API as long as the top- k order remains sufficiently stable.

4. Design

4.1. Overview

AsymMark-R, the rank instance of AsymMark, has four components, summarized in Fig. 1. First, the agent elicits or estimates P_t over candidate behaviors, following AgentMark's black-box-compatible behavior-distribution interface [6]. Second, the encoder computes the rank invariant by sorting behaviors and restricting embedding to the top- k region. Third, it applies the interleaved rank split from weakly asymmetric steganography [7] to behavior candidates rather than text symbols, preserving the behavior distribution while making the selected rank decodable. Fourth, deterministic RLNC maps the payload into a stream of coded bits so that the verifier can solve for the payload after observing enough independent coded bits.

4.2. Binary Distribution-Preserving Primitive

At a tree node, the encoder must choose between two groups with masses S_0 and S_1 while optionally embedding one bit. Assume $S_0 \geq S_1$ and $S = S_0 + S_1$. Given message bit $m \in \{0, 1\}$ and $r \leftarrow U[0, 1)$ from the DRBG, define

$$r_m = (rS + \frac{1}{2}Sm) \bmod S. \quad (4)$$

BinEnc selects group 0 if $r_m < S_0$ and group 1 otherwise. If group 1 is selected, one bit is embedded; if group 0 is selected, no bit is consumed and the same bit is retried at the next level. Because r_m is uniform over $[0, S)$ for uniform r , the selected group has exactly the desired marginal probabilities S_0/S and S_1/S .

Embedder: full channel	→	Trajectory log	→	Verifier: partial channel
Agent context h_t and candidate behaviors $\mathcal{B}_t$. Elicit probabilities P_t and sort top- k order σ_t . Use P_t to sample a watermarked behavior with the same marginal distribution.		Logged behavior $\hat{b}_t$, step context, claimed payload identifier, and audit metadata. The log may omit steps or be checked after model/API changes.		Reconstruct only the top- k order σ'_t from a restricted API or proxy model. Decode bits from the rank of $\hat{b}_t$; solve RLNC equations for the payload.

Figure 1. System view of AsymMark-R, the rank-order instance of AsymMark. The embedder needs calibrated probabilities to preserve utility; the verifier needs only the rank invariant for decoding, plus enough observed coded bits for RLNC recovery.

Why zero-bit branches are intentional. The primitive gives up a bit when the high-mass group is selected because forcing every branch to carry a bit would require reweighting the group probabilities. The deferred-bit design is the same kind of tradeoff used by distribution-preserving steganography: capacity is opportunistic, but the cover distribution is unchanged. In agent settings this is a feature rather than a cosmetic constraint, because utility regressions caused by shifted planning choices are easier for users and downstream systems to notice than small variations in textual style.

4.3. Rank Encoder

Let σ_t be the descending rank ordering of P_t . At step t , let $n_t = \min(k, |\mathcal{B}_t|)$ be the actual number of visible candidates in the top- k view. The encoder first samples whether to enter the visible region with probability $S_{\text{top}} = \sum_{i=1}^{n_t} P_t(\sigma_t(i))$. If not, it samples from the tail proportionally and embeds no bits. Inside the visible region, it normalizes probabilities and applies recursive splitting:

- 1) Split the current sorted list into even ranks $(0, 2, 4, \dots)$ and odd ranks $(1, 3, 5, \dots)$.
- 2) Use BinEnc on the two group masses.
- 3) Recurse into the selected group until a single behavior remains.

Interleaving is important. A top-half/bottom-half split would place most mass in the first group, producing low embedding opportunity. Even/odd splitting balances mass while preserving a rank path that the verifier can recover.

4.4. Rank Decoder

Given selected behavior $\hat{b}_t$ and top- k order σ_t , the decoder sets $n_t = \min(k, |\sigma_t|)$ and finds the zero-indexed rank q of $\hat{b}_t$ within the visible prefix. If $\hat{b}_t$ is outside that prefix, the step yields no bits. Otherwise, the decoder reconstructs the path by repeatedly reading the parity of q :

$$q_0 = q, \quad q_{\ell+1} = \lfloor q_\ell / 2 \rfloor. \quad (5)$$

Whenever q_ℓ is odd, the selected path entered the odd group and one bit was embedded. The value of that bit is determined by the same DRBG value used at level ℓ : the implementation decodes it as 1 if $r_\ell < 0.5$ and 0 otherwise. The decoder consumes exactly $\lceil \log_2 n_t \rceil$ DRBG values per step, matching the encoder’s fixed synchronization budget.

The common shorthand $\lceil \log_2 k \rceil$ is the saturated-candidate upper bound when $n_t = k$. Figure 2 summarizes the one-step encoder and decoder side by side.

4.5. Coding Layer

Per-step embedding capacity varies with P_t , k , and trajectory length. To avoid requiring a contiguous bitstream, AsymMark-R uses deterministic RLNC over GF(2), following the standard random linear network coding principle [13]. For payload $m \in \{0, 1\}^L$, the i -th coded bit is

$$c_i = \langle a_i, m \rangle \bmod 2, \quad (6)$$

where coefficient vector a_i is generated deterministically from a stream key and index i . The verifier collects decoded coded bits and solves the resulting linear system by Gaussian elimination over GF(2). This layer is inherited from AgentMark’s erasure-resilience design but is independent of whether the step decoder is symmetric or rank-only.

The coding layer also cleanly separates two questions that are often conflated. The step decoder asks whether an observed behavior can yield a reliable coded bit under the verifier’s channel view. RLNC asks whether enough independent coded bits have survived to solve for the payload. This separation is useful for weak asymmetry: a rank-only decoder may produce fewer bits than an exact-probability decoder, but the coding layer can still recover when the trajectory is long enough or the audit window spans enough steps.

4.6. Verification Protocol

An audit takes as input a pre-registered key/payload claim, a canonicalized trajectory log, and a verifier-side procedure for reconstructing top- k candidate orders. The verifier first derives the per-step DRBG streams from logged contexts and decodes all observed steps using RankDec. It then runs RLNC elimination over the recovered coded equations. If the system recovers the registered payload and the reconstructed ranks pass a deployment-specific quality gate, the audit reports a positive attribution. If the payload does not recover but the rank-quality gate fails or too few independent equations are available, the audit reports *inconclusive*. A negative attribution is reserved for the case where the verifier has enough rank-consistent evidence to

Encoder: uses probabilities to preserve policy**Input:** P_t , context h_t , key K_{sh} , coded stream M , top- k .

1. Derive $\mathcal{G}_t$ from (K_{sh}, h_t) .
2. Sort behaviors: $\sigma_t \leftarrow \text{argsort}(P_t)$; set $n_t = \min(k, |\mathcal{B}_t|)$.
3. Sample tail with probability $1 - \sum_{i \leq n_t} P_t(\sigma_t(i))$; tail emits no bits.
4. Inside top- n_t , normalize P_t and recursively split ranks into even/odd groups.
5. Run BinEnc on group masses; consume a coded bit only when the selected path enters the odd group.
6. Advance unused DRBG calls to the fixed budget $\lceil \log_2 n_t \rceil$.

Output: selected behavior $\hat{b}_t$ and consumed coded bits.

Shared alignment. Both sides share (K_{sh}, h_t) and the fixed DRBG budget; only the encoder needs calibrated probabilities. Correct decoding requires the verifier to reconstruct the same top- k rank path, after which RLNC pools decoded bits across the audit window.

Decoder: uses ranks, never probabilities**Input:** $\hat{b}_t$, verifier top- k order σ'_t , context h_t , key K_{sh} .

1. Set $n_t = \min(k, |\sigma'_t|)$ and derive the same DRBG sequence.
 2. If $\hat{b}_t \notin \sigma'_t[1 : n_t]$, return no bits.
 3. Let q be the zero-indexed rank of $\hat{b}_t$ in σ'_t .
 4. Reconstruct the path by parity: odd q means the encoder entered the odd group and emitted one bit.
 5. Decode that bit from the corresponding DRBG draw; set $q \leftarrow \lfloor q/2 \rfloor$ and repeat.
 6. Consume exactly $\lceil \log_2 n_t \rceil$ DRBG calls, matching the encoder.
- Output:** decoded coded bits for step t .

Figure 2. One-step rank codec. The encoder uses probability masses only to sample from the original behavior policy, while the verifier decodes from the selected behavior’s top- k rank path.

decode but the recovered payload contradicts the registered claim.

The rank-quality gate is intentionally separate from the watermark statistic. A verifier can estimate it using held-out calibration contexts, duplicate model queries, or agreement between independent rankers. This prevents a common operational error: treating a failed rank-only decode as evidence that a trajectory is clean when the verifier’s channel reconstruction is itself unreliable.

Before an audit window has accumulated enough independent RLNC equations, the verifier can still estimate whether the recovered equations are informative under a binomial null model of random agreement. This diagnostic can trigger further collection or manual review, but it is not by itself a positive attribution decision.

Appendix A expands the distribution-preservation proof, Appendix B details the fixed-budget synchronization rule, and Appendix C summarizes how decoded bits become payload equations. These appendices are not separate claims; they make the construction auditable without interrupting the main channel-knowledge argument.

The top- k parameter controls a reliability–capacity tradeoff. Larger k exposes more rank information, but it also asks the verifier to reconstruct lower-probability alternatives whose order may be less stable across models and serving conditions. Smaller k is more conservative and better aligned with restricted APIs, but it leaves less room for embedding. We therefore evaluate a sweep rather than tuning to a single universal value; a deployment should choose k from calibration data by maximizing expected payload recovery subject to a rank-agreement threshold.

5. Security Analysis

Proposition 1 (distribution preservation). For each step t and behavior $b \in \mathcal{B}_t$, AsymMark-R outputs b with probability $P_t(b)$.

Proof sketch. The top/tail gate selects the visible top- n_t region with probability equal to its total mass and otherwise samples the tail proportionally. Tail samples emit no coded bits, so the verifier need not reproduce the gate randomness. Conditioned on entering the visible region, each binary split

uses BinEnc, whose selected group probability equals its normalized mass because r_m is uniform. Induction over the partition tree gives each leaf its normalized probability within top- n_t . Multiplying by the visible-region mass recovers $P_t(b)$. $\square$

Lemma 1 (probability-bin instability). There exist two behavior distributions with identical rank order for which a probability-dependent bin decoder assigns the same selected behavior to different bins.

Proof sketch. Consider four behaviors ordered $a \succ b \succ c \succ d$ and a two-bin prefix decoder whose first bin is the smallest rank prefix whose cumulative probability exceeds $1/2$. Under $P = (0.45, 0.30, 0.15, 0.10)$, the first bin is $\{a, b\}$. Under $Q = (0.55, 0.25, 0.12, 0.08)$, the rank order is unchanged but the first bin is $\{a\}$. The selected behavior b is therefore interpreted in different bins although the verifier preserved the full ranking. Differential recombination uses a richer bin structure than this toy prefix rule, but it shares the same dependence on probability masses; rank agreement alone does not fix its decoder partition. $\square$

Proposition 2 (rank-noise robustness boundary). Under an adjacent-swap noise model with per-adjacent-pair swap probability ϵ , suppose a selected rank’s decoded path of depth $d_t = \lceil \log_2 n_t \rceil$ can be corrupted by at most cd_t relevant local swaps, where $n_t = \min(k, |\mathcal{B}_t|)$ is the actual visible candidate count. Then the per-step path-corruption probability is bounded by

$$p_{\text{flip},t} \leq 1 - (1 - \epsilon)^{c \lceil \log_2 n_t \rceil}. \quad (7)$$

Combining this bound with the coded-evidence model below gives the observation budget needed to distinguish the watermark from the null at a chosen tail probability. The bound is deliberately model-dependent: arbitrary targeted rank manipulation remains outside the weak-asymmetric guarantee, but the expression captures the empirical tradeoff that larger k gives more path bits while exposing more rank positions to noise. When all steps fill the requested top- k view, $\lceil \log_2 k \rceil$ is the corresponding upper-bound depth.

Corollary 1 (zero-noise rank preservation). Let P_t and P'_t satisfy $\text{argsort}_k(P_t) = \text{argsort}_k(P'_t)$. For any selected behavior $\hat{b}_t$ and shared key/context-derived DRBG stream, RankDec returns the same bits under both distributions. This

follows immediately from Proposition 2 with $\epsilon = 0$: the decoder accesses channel information only through the rank position of $\hat{b}_t$, and the DRBG stream depends on the key and context rather than probability values.

Corollary 2 (coded-bit evidence under rank agreement). Let ρ be the probability that an observed embedding step is top- k rank-consistent between embedder and verifier. Suppose rank-consistent steps decode a coded bit correctly with probability $q > 1/2$, while rank-inconsistent steps are independent of the claimed bit. Then the expected coded-bit agreement rate is

$$\mathbb{E}[\bar{X}] = \frac{1}{2} + \rho(q - \frac{1}{2}). \quad (8)$$

For N decoded bits, Hoeffding's inequality gives

$$\Pr\left[\bar{X} \leq \frac{1}{2} + \gamma\right] \leq \exp\left(-2N\left(\rho(q - \frac{1}{2}) - \gamma\right)^2\right) \quad (9)$$

whenever $0 < \gamma < \rho(q - \frac{1}{2})$. Thus lower rank agreement does not create a hard impossibility; it increases the evidence budget needed to distinguish the watermark from the null. This corollary is a diagnostic model for coded-bit evidence, not a replacement for pooled payload recovery.

Proposition 3 (packet scarcity). For an L -bit deterministic RLNC payload, recovery over an audit window requires at least L independent coded packets. If M independent packets survive after top- k misses, erasures, and rank failures, then

$$\Pr[\text{recover}] \leq \Pr[M \geq L]. \quad (10)$$

For an audit window pooling trajectories $i = 1, \dots, n$, the expected packet budget is additive:

$$\mathbb{E}[M_{\text{pool}}] = \sum_i \mathbb{E}[M_i]. \quad (11)$$

If each trajectory has expected packet yield $\mu_{\text{traj}} = N(1 - r)\text{DSR}(k, \epsilon)$ under erasure rate r and rank-noise level ϵ , then a deployment aiming for L independent equations plus a rank-deficiency margin δ needs, in expectation,

$$n \gtrsim \frac{L + \delta}{\mu_{\text{traj}}}. \quad (12)$$

This explains why short ToolBench traces can provide useful packet evidence while requiring cumulative audit windows for payload recovery: pooling trajectories converts sparse per-trace packets into enough independent equations.

Proposition 4 (top- k marginal gain and saturation). Let $C_{\text{nom}}(k)$ be the nominal coded-bit opportunity exposed by a top- k verifier view, and let $\text{DSR}(k)$ be the observed decode success rate under that view. Increasing k can raise $C_{\text{eff}}(k) = C_{\text{nom}}(k)\text{DSR}(k)$ in two ways: by exposing more coded bits, and in packet-limited regimes by pushing more trajectories over the minimum packet threshold for payload recovery. Once additional ranks mostly add unstable low-probability candidates, marginal gains should saturate and may turn negative. This statement does not imply an internal optimum in every measured range. If the measured k values

do not reach the regime where rank instability dominates, $C_{\text{eff}}(k)$ can be monotone over the tested interval and the best observed k is only a lower bound on the true operating point.

Minimum evidence threshold. For bit-level triage before pooled RLNC recovery, a one-sided binomial test with match probability $q > 1/2$ under the watermark and $1/2$ under the null needs on the order of

$$N_{\min} = O\left(\frac{z_\alpha^2}{(q - 1/2)^2}\right) \quad (13)$$

observed decoded bits to reach significance level α . This threshold is not an attribution decision by itself; it sizes the observation window needed before a verifier should expect enough channel evidence to justify attempting pooled payload recovery.

False positives. For an unwatermarked or unrelated audit window, positive attribution requires recovery of a key and payload claim that was registered before inspection. Under the standard null that decoded clean bits are independent of this fixed claim and that no adaptive post-hoc key or payload search is counted as a single test, an L -bit payload match occurs with probability 2^{-L} . In practice the verifier can tune L to the desired audit threshold.

Why the theory matches the contribution. Lemma 1 gives the negative result: rank agreement alone is insufficient for probability-bin decoders. Proposition 2 is the quantitative robustness statement for AsymMark-R under rank noise, with Corollary 1 giving the zero-noise rank-preservation case. Proposition 3 separates channel robustness from evidence volume, which is why the evaluation reports payload recovery as pooled audit-window recovery rather than isolated trajectory success. Proposition 4 explains why top- k should be calibrated rather than treated as a universal constant: larger k can raise nominal capacity, but the measured optimum is only meaningful once the sweep reaches the regime where rank instability erodes decode success.

Key privacy. The deployment watermark key is used only to derive context-specific pseudorandomness. The public trajectory and rank list do not reveal the key under the pseudorandom function assumption for HMAC-SHA512. A verifier may check many trajectories under the same registered owner only with domain-separated contexts, key identifiers, and tenant/product isolation; operational deployments should rotate keys to limit blast radius if a key is disclosed. The offline experiments use recorded payload metadata and deterministic replay to evaluate the watermark channel; they should not be read as evidence that a public context-derived prototype default provides secret-key security.

Claim binding. A watermark match is meaningful only for a claimed key and payload that were bound to an owner before the audit. Otherwise, an operator could test many keys or payloads after observing a trajectory and inflate the effective false-positive rate. Deployments should therefore register a key identifier, payload namespace, and issuance time in an append-only registry or signed manifest. The

verifier can then count only pre-registered claims and apply the multiple-testing correction in Section 6.11.

Limits. AsymMark-R is robust to value perturbations that preserve top- k order, not to arbitrary rank manipulation. If an adversary can force targeted rank swaps in the top- k list, it can corrupt decoded paths. This is the expected boundary of any rank-only weakly asymmetric scheme. The evaluation therefore includes a rank-disagreement proxy and treats live cross-model rank reconstruction as deployment work. Appendix D states the corresponding rank-stability intuition and the resulting top- k choice rule.

6. Evaluation

6.1. Setup

We evaluate on ALFWorld [8] and ToolBench [9]. ALFWorld tests embodied planning over in-distribution (ID) and out-of-distribution (OOD) splits; ToolBench tests tool-use behavior across six split categories. These are the same kinds of behavior-level settings used by AgentMark to establish utility-preserving behavioral watermarking; here they serve a narrower purpose: holding the behavior-watermark carrier fixed while changing the verifier’s channel knowledge. The LLM endpoints are DeepSeek Chat (deepseek-chat) [14] and Gemini 2.0 Flash (gemini-2.0-flash) [3]. The postprocessed corpus contains 11,820 trajectories and 4,728 watermark decode-capable trajectories. Methods are: vanilla LLM agent, clean AgentMark pipeline without embedding, red-green (RG) behavior bias, symmetric AgentMark-F, and rank-based AsymMark-R. RG follows the Agent Guide style of keyed behavior steering [15]; it is included as a high-level contrast to distribution-preserving watermarking rather than as a tuned competitor.

Unless otherwise stated, recovery rates are computed from the available offline decoder artifacts. Capacity tables are bit-level channel proxies over logged trajectories and should not be conflated with deployment attribution. Payload recovery is reported as cumulative audit-window recovery: packets from trajectories in the same dataset/model/split/run/configuration are combined before RLNC decoding. This matches the pooled recovery curve in Fig. 3 and treats logged trajectories as packet sources for the same cumulative audit window. Deployment attribution should use registered payloads sized to the required false-positive threshold and the false-positive accounting in Section 6.11.

Evaluation coverage after postprocessing is balanced across the two watermarking methods. ALFWorld contributes four model/split groups with 3/3 completed runs per group and 3,288 decode-capable trajectories, split evenly as 1,644 per watermark. ToolBench contributes two model/split groups with 18/18 completed runs per group and 1,440 decode-capable trajectories, split evenly as 720 per watermark. Utility and stealth metrics use the available methods, with vanilla reported separately in Appendix I; decode-capable counts apply only to the two watermarking methods

because vanilla, clean, and RG trajectories do not carry payload metadata.

6.2. Protocol

The evaluation pipeline starts from logged trajectories containing the selected behavior, candidate list, behavior probabilities when available, and watermark metadata. Utility and JSD are computed over all valid task trajectories. Watermark recovery is computed on the decode-capable subset: trajectories with a recorded watermark configuration, a recoverable candidate ordering, and enough logged context to replay the deterministic DRBG schedule. This distinction is reported explicitly in the corpus counts above.

For each decode-capable trajectory, we run the same verifier under several channel views. The *exact* view uses the logged probabilities as a diagnostic upper bound rather than as the target audit interface. The *top- k* views discard numeric probabilities and retain only the ordered candidate list. The selected-only view is used as a negative control when available. Rank-noise experiments use a bit-level disagreement proxy parameterized by ϵ because the postprocessed logs do not contain full alternate rank lists for every verifier; erasure experiments remove a random fraction of observed steps before RLNC solving. All tables aggregate over the available model/split groups rather than selecting the best configuration for each task. The main rank-only comparison reports fixed top-10, top-6, and top-2 views; the separate top- k ablation reports the full sweep and is not used to choose per-model winners. These aggregates are descriptive comparisons over the generated corpus; they are not presented as hypothesis tests for small differences between methods.

6.3. Reproducibility

The evaluation package is organized so that each reported number is tied to a reproducible postprocessing artifact. The main result package supports utility, stealth, verifier-channel capacity, pooled recovery, and matched top- k recovery; supplemental artifacts cover rank noise, erasure, live cross-model reranking, semantic rewrites, evidence-strength diagnostics, and false-positive checks. Appendix J summarizes this mapping at the claim level.

6.4. Evaluation Questions

The evaluation follows the weak-asymmetry story rather than a table-by-table result dump. First, after inheriting AgentMark’s distribution-preserving behavior channel, does the rank instance preserve task utility and behavior-frequency statistics? Second, what happens when an exact-channel baseline is evaluated through the weaker rank-only verifier channel? Third, what reliability–efficiency tradeoff appears as the verifier changes top- k depth and audit-window size? Boundary diagnostics for rank noise, erasure, semantic rewrites, cross-model reranking, and bit-level evidence thresholds are reported in the appendices so that the main text stays focused on the verifier-channel comparison.

6.5. Metrics

We report task success rate and average steps for utility. We report Jensen-Shannon divergence (JSD) between empirical action-frequency distributions for a coarse stealth check. For watermark recovery, *payload recovery* is the fraction of pooled audit windows whose decoded payload exactly matches the embedded payload under the payload convention defined in the setup. When a verifier view does not recover the registered payload in a window, its payload recovery for that window is zero. We also report proxy C_{eff} in units of effective proxy bits per task trajectory:

$$C_{\text{eff}}^{\text{proxy}} = C_{\text{nom}}^{\text{proxy}} \cdot \text{DSR}^{\text{proxy}}. \quad (14)$$

Here $C_{\text{nom}}^{\text{proxy}}$ is the mean decoded proxy-bit contribution from each logged trace, and $\text{DSR}^{\text{proxy}}$ is proxy payload recovery over the pooled audit window. Thus $C_{\text{eff}}^{\text{proxy}}$ is a per-trajectory effective-capacity diagnostic; a per-step version would divide it by the mean trajectory length. This score is useful for comparing channel views because a high-capacity exact-channel scheme that fails under rank-only verification is scored as low usable capacity in that verifier channel. For confidence analysis, we use a one-sided binomial tail probability under the null hypothesis that each decoded coded bit agrees with the claim with probability $1/2$.

6.6. Verifier-Channel Diagnosis

The central empirical comparison mirrors the weak-asymmetric tradeoff. Exact probabilities are the native channel for AgentMark-F, and Table 1 shows that AgentMark-F has higher exact-channel C_{eff} on ALFWorld. The question is whether that advantage survives the weaker verifier interface that motivated this paper. The $L1$ capacity column and the channel-ladder figure test this mechanism. If a decoder depends on probability bins, Lemma 1 predicts that preserving candidate order is not enough to preserve decoder state. If a decoder depends only on rank path, Corollary 1 predicts no loss from probability-value perturbations as long as ranks are preserved. We therefore treat the exact-probability view as an upper-bound capacity diagnostic, the selected-only view as a lower-bound negative control, and the top- k rank view as the partial-knowledge target.

6.7. Utility and Behavioral Stealth

Table 1 summarizes the main result along the three quantities needed for the channel-knowledge story: task utility, behavior-frequency stealth, and effective capacity under exact versus rank-only verifier views. On ALFWorld, AsymMark-R reaches 78.6% success versus 80.0% for clean and 79.4% for AgentMark-F. On ToolBench, AsymMark-R reaches 77.1% versus 75.7% for the clean pipeline. We interpret these aggregates conservatively: rank encoding stays close to the clean and symmetric pipelines and does not exhibit the large utility tax associated with naive behavior bias, but small one- or two-point differences are

not treated as significant. The capacity columns are shown only for payload-carrying methods and report $C_{\text{eff}}^{\text{proxy}}$ per task trajectory. The retained column is the ratio $L1/L0$, making the verifier-channel degradation explicit; it should be read together with the absolute $L1$ capacity because high retention on a low-capacity domain does not imply higher usable capacity.

6.8. Rank-Only Verification

Under exact probabilities on ALFWorld, AgentMark-F has higher payload recovery (0.700) than AsymMark-R (0.532), reflecting the efficiency cost of rank-only structure. Under top-10 rank-only verification, AgentMark-F collapses to 0.014 payload recovery, while AsymMark-R remains at 0.068. In C_{eff} terms, ALFWorld AgentMark-F falls from 19.692 under exact probabilities to 0.333 under top-10 ranks, while AsymMark-R falls from 7.707 to 1.000. Figure 4 plots the same ladder as relative retained capacity, normalizing each method/domain series by its own $L0$ value so that both benchmark panels show the degradation shape while the absolute C_{eff} values remain in Table 1.

We also add the step-level counterfactual suggested by Lemma 1. For each AgentMark-F step whose exact decoder emits bits, we keep the top- k order fixed, drop candidates outside the verifier’s top- k view, renormalize the remaining probabilities, and re-run the same context-derived decoder on the same selected behavior. This is a truncation-and-renormalization instance of the value perturbations in Lemma 1, not an exhaustive test of all possible calibrated-probability changes. When the selected behavior remains inside top- k , any changed bin or changed bits are therefore probability-bin effects rather than rank-order effects; the bin-change and bit-change percentages below are conditional on that selected-in-top- k event, while selected-dropped rates are measured over exact-decodable steps. On ALFWorld DeepSeek, among rank-stable steps, top-10 still changes the probability bin in 13.7% of steps and flips decoded bits in 11.1%; top-4 changes bins in 56.8% and flips bits in 51.3%. On ToolBench DeepSeek, top-10 is milder but nonzero (4.6% bin changes, 3.7% bit changes), while top-4 again exposes a large bin-instability regime (52.6% bin changes, 43.6% bit changes). This turns the earlier aggregate observation into a direct per-step attribution: exact-channel probability bins are unstable even when the selected behavior and retained top- k rank order are unchanged.

ToolBench is lower-capacity for both schemes because trajectories are short and many tool choices have low recoverable entropy. Nevertheless, AsymMark-R keeps nearly the same payload recovery from exact probabilities to top-10 rank-only decoding (0.033 to 0.032), whereas AgentMark-F drops from 0.085 to 0.003 payload recovery. The same pattern appears in C_{eff} : ToolBench AgentMark-F drops from 0.248 to 0.009, while AsymMark-R remains effectively flat from 0.053 to 0.050.

AsymMark-R weakens exact probability knowledge to rank knowledge; it does not claim attribution from the action log alone. A selected-behavior-only verifier recovers

TABLE 1. MAIN UTILITY, STEALTH, AND VERIFIER-CHANNEL CAPACITY SUMMARY. SUCCESS, STEPS, JSD, AND $C_{\text{eff}}/\text{TASK}$ REPORT MEAN $\pm$ STANDARD DEVIATION OVER THE AVAILABLE AGGREGATION CELLS; LOWER JSD IS CLOSER TO CLEAN BEHAVIOR FREQUENCIES. $L0$ IS EXACT-PROBABILITY VERIFICATION, $L1$ IS TOP-10 RANK-ONLY VERIFICATION, $C_{\text{eff}}/\text{TASK}$ IS THE PROXY EFFECTIVE CAPACITY PER TASK TRAJECTORY, AND RETAINED IS THE RATIO OF MEANS $L1/L0$. RG DENOTES RED-GREEN BEHAVIOR BIAS.

Dataset	Method	SR (%) $\uparrow$	Steps $\downarrow$	JSD $\downarrow$	$C_{\text{eff}}/\text{task}$ $L0$ $\uparrow$	$C_{\text{eff}}/\text{task}$ $L1$ $\uparrow$	Retained $L1/L0$ $\uparrow$
ALFWorld	Clean	80.0 $\pm$ 10.5	16.23 $\pm$ 2.19	0.000	—	—	—
	RG	74.9 $\pm$ 13.1	17.58 $\pm$ 2.75	0.134 $\pm$ 0.061	—	—	—
	AgentMark-F	79.4 $\pm$ 10.7	16.58 $\pm$ 1.89	0.122 $\pm$ 0.079	19.692 $\pm$ 8.109	0.333 $\pm$ 0.168	1.7%
	AsymMark-R	78.6 $\pm$ 10.0	16.40 $\pm$ 2.11	0.121 $\pm$ 0.070	7.707 $\pm$ 4.590	1.000 $\pm$ 0.730	13.0%
ToolBench	Clean	75.7 $\pm$ 2.4	5.32 $\pm$ 0.78	0.000	—	—	—
	RG	75.1 $\pm$ 3.2	4.78 $\pm$ 0.01	0.181 $\pm$ 0.069	—	—	—
	AgentMark-F	74.7 $\pm$ 0.2	4.54 $\pm$ 0.04	0.174 $\pm$ 0.077	0.248 $\pm$ 0.182	0.009 $\pm$ 0.020	3.6%
	AsymMark-R	77.1 $\pm$ 0.5	5.07 $\pm$ 0.77	0.140 $\pm$ 0.058	0.053 $\pm$ 0.038	0.050 $\pm$ 0.036	94.3%

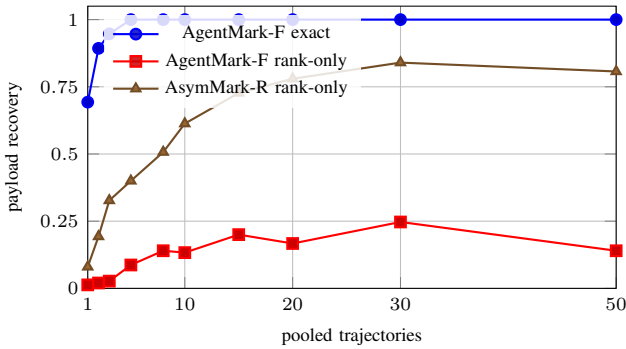


Figure 3. Rank-only pooled recovery on ALFWorld/DeepSeek/ID, top-10 verifier view. The plotted window stops at 50 trajectories to show the audit-window regime before very large pools require conflict-aware aggregation.

no payload in these diagnostics, so we keep it as a negative-control sanity check rather than a main result column.

The same mechanism appears more clearly in pooled audit windows. Figure 3 compares ALFWorld/DeepSeek/ID top-10 recovery as the verifier increases the pooled trajectory budget within the same audit configuration. AgentMark-F reaches perfect recovery by pool size 5 when exact probabilities are available, confirming that the baseline is not packet-limited in its native channel. Under a rank-only surrogate that keeps only the top-10 order and assigns monotone linear pseudo-probabilities before running the AgentMark-F differential decoder, pooling does not rescue the baseline: recovery remains between 0.013 and 0.247 over the plotted window. In contrast, AsymMark-R rises to 0.613 at pool size 10, 0.780 at 20, and 0.807 at 50. The point is not that AsymMark-R dominates exact-channel AgentMark-F; it is that the rank-only decoder matches the verifier channel, while a probability-bin decoder continues to accumulate conflicting equations even when the audit window grows.

6.9. Top- k Calibration

Increasing k raises both the amount of available rank information and the risk of rank instability, matching Proposition 4’s saturation view. Figure 5 reports the matched-

top- k ToolBench reruns where the embedder and verifier use the same rank depth and RLNC recovery is evaluated on actual generated trajectories. Single trajectories almost never recover the 8-bit payload because ToolBench traces are short and packet yield is sparse. Pooling trajectories into larger audit windows shows the operating regime directly: very small k can be packet-limited, while moderate and larger k provide enough independent packets for same-model recovery.

This calibration is a real decoding result, not the offline capacity proxy used for Table 1. We keep the ALFWorld top- k sweep out of this figure because the local ALFWorld artifact is an offline replay over logged rank traces rather than a matched-top- k rerun. The corresponding proxy still helps explain candidate-set saturation, but it should not be read as task-level or audit-window payload recovery.

6.10. Supplemental Boundary Checks

The remaining experiments check where the rank channel starts to fail rather than defining the main contribution. Appendix E reports adjacent-swap rank-noise and step-erasure diagnostics; Appendix F reports semantic rewrite stress tests; Appendix G reports live cross-model ToolBench reranking. Together they support the same operating rule used above: choose the verifier view by expected usable packet yield, not by nominal rank depth alone. When the verifier preserves the rank path, AsymMark-R decodes from that invariant; when rank paths drift, the audit should increase the evidence window, reduce k , or report an inconclusive result.

6.11. False Positives and Evidence Thresholds

False-positive control is analytic for exact payload claims. If a clean or unrelated audit window yields coded bits that are independent of a fixed pre-registered payload claim, and no adaptive post-hoc key or payload search is counted as the same test, the chance that it recovers that exact L -bit payload is 2^{-L} . Operational deployments should choose registered payload lengths according to the required audit threshold and available trajectory length.

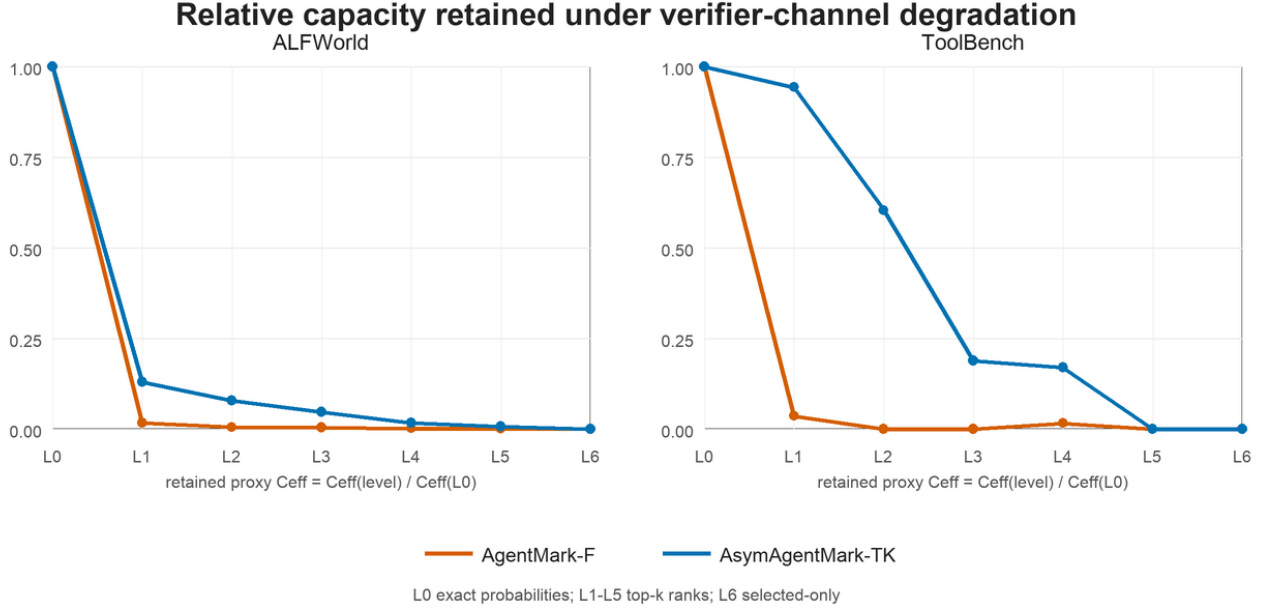


Figure 4. Relative effective capacity retained over the L0–L6 verifier-channel ladder. The two panels separate ALFWorld and ToolBench because their absolute proxy C_{eff} scales differ by almost two orders of magnitude; each curve is normalized by its own L0 exact-probability value. AgentMark-F loses usable capacity as soon as the verifier only has ranks, while AsymMark-R degrades more gradually under top- k rank views.

The threshold must also account for multiple testing. If an operator tests N independent key/payload claims against the same corpus, a conservative union bound gives a family-wise false-positive rate at most $N2^{-L}$. This is why payload length is an audit-policy parameter rather than an implementation detail: larger claim spaces are needed when many owners, keys, or payloads may be tested against the same corpus.

Empirical false-positive trials follow this analytic curve across the tested datasets, models, and top- k configurations. Because the attribution event is exact payload equality for a pre-registered claim, we report the analytic per-claim threshold 2^{-L} as the security control and use random-claim trials only to check that the implementation behaves like the null model. Appendix J maps this and the other reported diagnostics at the claim level.

Bit-level scores require a different interpretation. When a verifier reports a match rate over n decoded coded bits rather than a recovered payload, the appropriate null is a binomial tail $\Pr[X \geq s; X \sim \text{Binom}(n, 1/2)]$ for the observed score s . This statistic is useful for sizing evidence windows before enough independent RLNC equations are available; Appendix H gives the corresponding confidence curves. In the main protocol, positive attribution is the recovered pre-registered payload, not an isolated bit-level score.

7. Discussion

Exact probability is a luxury. If exact P_t is available, symmetric AgentMark-F can be more capacity-efficient. If the verifier has only top- k ranks, symmetric decoding be-

comes unreliable, and a rank-decodable construction is the relevant design point. ToolBench further shows the packet-scarce regime: short tool-use traces can preserve utility while still requiring pooled audit windows for payload recovery.

Calibration and limits. Rank consistency is easier to measure than exact probability equality: a verifier can report top- k agreement, Kendall tau distance, or adjacent-swap rates before interpreting a failed decode. AsymMark-R is robust to value perturbations that preserve the rank path, not to arbitrary rank manipulation or unreliable candidate canonicalization. A practical verifier should therefore distinguish negative attribution from inconclusive audits caused by poor rank reconstruction or too few independent equations.

Framework scope. AsymMark is defined by the separation between a policy-preserving embedding channel and a verifier-reconstructible decoding invariant. Top- k rank order is the invariant we implement and evaluate because it is naturally exposed by restricted agent APIs and sufficient to recover the interleaved partition path. Other invariants, such as coarse probability buckets or keyed canonical orders over candidate sets, could instantiate the same template when they are more stable for a deployment.

Deployment path. The results establish partial-channel feasibility rather than turnkey deployment certification. A production audit should add live end-to-end RLNC recovery over the deployed logging stack, calibrated rank reconstruction for the deployed verifier model or API, and pre-registered decision thresholds. This does not change the contribution: when calibrated probabilities cannot be trusted but top- k ranks can, decode from the invariant that survives.

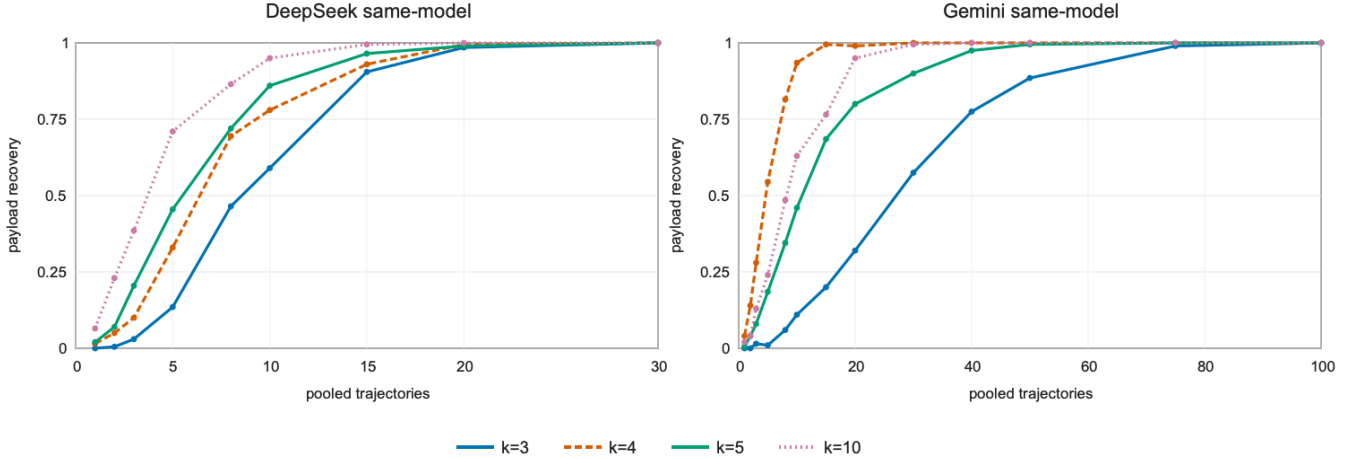


Figure 5. ToolBench matched-top- k same-model real decoding calibration for AsymMark-R. Curves show sampled pooled payload recovery for $k \in \{3, 4, 5, 10\}$ as the audit window grows; each panel uses trajectories generated and verified by the same model, with panel-specific x-axis ranges to show the different saturation regimes.

8. Related Work

LLM and agent watermarking. Token-level LLM watermarking modifies or tests generated text distributions [4], [5], [16], [17], but agents often compile plans into structured actions where final text no longer carries the decision signal. Agent Guide biases agents toward keyed behavior subsets [15]; AgentMark [6] instead preserves per-step behavior distributions and is the direct predecessor of this work. It evaluates multi-bit behavioral identifiers, utility preservation, partial-log recovery under erasure and truncation, semantic-preserving observation rewriting, and compatibility with action-layer content watermarks. AsymMark-R treats those results as the exact-channel starting point and asks an orthogonal question: what verifier channel is needed to decode a distribution-preserving behavioral payload when the original probability list cannot be replayed?

Steganography and weak asymmetry. Provably secure steganography formalizes indistinguishability between cover and stego content [11]; Meteor and later language-model steganography improve practical distribution-preserving sampling [12], [18], [19], [20]. Weakly asymmetric steganography studies receivers with partial channel knowledge and gives the rank construction used by AsymMark-R [7]. That work evaluates text steganography through capacity, correctness, and numerical distribution mismatch; our contribution is the behavior-channel framework, audit protocol, synchronization, RLNC integration, and agent-trajectory evaluation around the same partial-knowledge principle.

Coding and benchmarks. Random linear network coding recovers payloads from sufficiently independent subsets of coded symbols [13], matching the partial-log nature of agent trajectories. ALFWorld [8], ToolBench [9], and OASIS [10] illustrate why behavior-level provenance matters: the action trajectory can be more consequential than the final message.

9. Threats to Validity and Ethics

Offline postprocessing. Capacity and robustness experiments are computed from logged trajectories and offline decoder proxies, with matched top- k and live rerank studies used as calibration evidence. This setup isolates the verifier-channel question, but deployment use should still validate end-to-end RLNC recovery under the production logging stack.

Rank reconstruction and benchmarks. The theory applies when the verifier preserves the top- k rank path. ALFWorld and ToolBench cover embodied planning and tool use, but other agent domains may differ in rank stability, trajectory length, and candidate-set structure. The best k , payload length, and rank-quality threshold should therefore be calibrated per deployment.

Canonicalization and elicitation. Rank decoding assumes that the logged behavior and verifier-side candidates share a canonical behavior representation. This is straightforward for enumerated actions and structured tool calls, but less automatic for free-form plans, GUI operations, or APIs whose schemas evolve. The method also assumes that the embedder elicits a meaningful behavior distribution; distribution preservation protects that elicited channel.

9.1. Ethics Considerations

This work is intended for provenance, audit, and authorized ownership verification. The evaluation uses benchmark task trajectories rather than private user data, but real deployments should treat trajectory logs as sensitive operational records, minimize retained context, and bind verification to an explicit audit purpose. Because overconfident attribution can cause harm, deployments should publish thresholds, report inconclusive outcomes, retain audit records, and separate watermark evidence from broader incident investigation.

10. Conclusion

Behavioral provenance for LLM agents should not depend on a verifier reproducing exact per-step probabilities. This paper identifies probability-bin instability as a verifier-channel failure mode for symmetric behavioral watermarking and formulates AsymMark, a weakly asymmetric framework that decouples the policy-preserving embedding channel from a verifier-reconstructible decoding invariant. Its rank instance, AsymMark-R, shows that top- k rank order can retain useful watermark signal while preserving the agent’s marginal behavior policy. Exact-channel AgentMark-F remains attractive when original probabilities are available; AsymMark-R targets audits where ranks survive and probabilities do not.

Appendix A. Distribution-Preservation Details

This appendix expands the proof sketch from Section 5. At any internal node of the top- k partition tree, let the even-rank group have normalized mass S_0 and the odd-rank group have mass S_1 , with $S_0 + S_1 = S$. Assume without loss of generality that $S_0 \geq S_1$; otherwise the group labels are swapped for the binary primitive. For $r \leftarrow U[0, 1)$ and message bit $m \in \{0, 1\}$, BinEnc computes

$$r_m = (rS + \frac{1}{2}Sm) \bmod S.$$

For either fixed m , r_m is uniform over $[0, S)$ because adding a constant modulo S preserves uniformity. Therefore the probability of selecting group j is exactly S_j/S . When the higher-mass group is selected, no bit is consumed; when the lower-mass group is selected, one bit is consumed. Bit consumption changes the message pointer but not the marginal group-selection probability.

Inducting over the recursive partition tree, every visible leaf b is selected with probability equal to its normalized top- n_t mass $P_t(b)/S_{\text{top}}$. The top/tail gate enters the tree with probability $S_{\text{top}} = \sum_{i < n_t} P_t(\sigma_t(i))$ and otherwise samples the tail proportionally. Hence every behavior, including tail behaviors that carry no bits, has marginal probability $P_t(b)$.

Appendix B. DRBG Synchronization

The encoder and decoder derive a per-step HMAC-SHA512 DRBG stream from the shared key and logged context. The important invariant is not the exact digest format, but the fixed consumption budget. For a top- k step with actual visible candidate count $n_t = \min(k, |\mathcal{B}_t|)$, both sides reserve

$$B_t = \lceil \log_2 n_t \rceil$$

DRBG draws for the rank tree. The encoder may terminate early when a singleton leaf is reached or when a zero-bit branch defers the message bit, but it still advances the stream to consume exactly B_t draws. The decoder independently

consumes the same B_t draws before returning the recovered bits. The commonly reported $\lceil \log_2 k \rceil$ budget is the special case in which the step fills the requested top- k view. Tail selections emit no bits and are treated as erasures by the verifier.

This fixed-budget rule prevents a local zero-bit event from shifting every later step. Without it, the decoder would have to know whether the encoder consumed a message bit or terminated early at each level, which is precisely the information it is trying to infer from the rank path.

Appendix C. RLNC Payload Recovery

The rank decoder outputs a variable number of coded bits from each observed step. Deterministic RLNC converts these bits into payload equations. For an L -bit payload m , the i -th coded bit is $c_i = \langle a_i, m \rangle \bmod 2$, where a_i is generated from a deterministic coefficient stream. The verifier collects decoded pairs (a_i, c_i) and solves the resulting linear system over GF(2). Recovery succeeds when the collected coefficient matrix has rank L and the decoded right-hand side is consistent.

This layer is agnostic to whether a coded bit came from exact-probability decoding or rank-only decoding. Rank-only verification typically produces fewer reliable equations, so it may need larger audit windows. The capacity diagnostics in the main text use this layer to compare verifier channels; deployment identifiers require longer registered payloads and live end-to-end recovery over pooled logs.

Appendix D. Rank Stability and Top- k Choice

Temperature rescaling preserves rank order because $p_i^{1/T}/Z$ is monotone in p_i for $T > 0$. Quantization, model refreshes, and proxy re-querying need not preserve ranks exactly, but adjacent swaps are most likely when score gaps are small. This motivates top- k restriction: high-probability behaviors usually have larger gaps and more stable semantic alternatives than long-tail candidates.

The top- k parameter therefore trades nominal capacity against reconstruction risk. Larger k gives more leaves and potentially more decoded bits per step, but it asks the verifier to reconstruct lower-probability ranks. Smaller k is conservative and compatible with restricted APIs, but it may underuse high-entropy trajectories. The deployment choice should maximize calibrated $C_{\text{eff}}(k)$, not raw decoded length: a deeper top- k view is useful only when its marginal capacity gain exceeds the marginal decode-success loss. The evaluation reports a sweep rather than selecting a universal k because the best choice is environment-dependent.

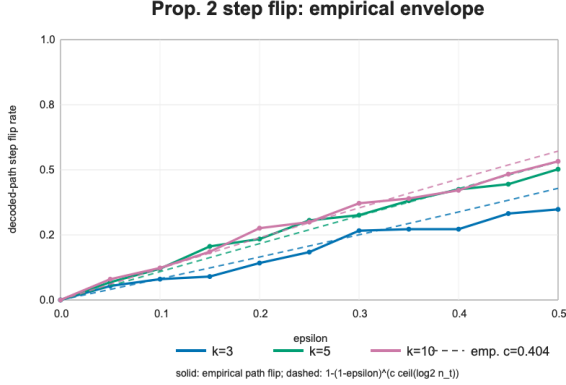


Figure 6. Fine-grid ToolBench rank-noise diagnostic for Proposition 2. Solid lines show empirical decoded-path step-flip rates under adjacent swaps; dashed lines show an empirical envelope $1 - (1 - \epsilon)^{c \lceil \log_2 n_t \rceil}$ with fitted $c = 0.404$.

Appendix E. Robustness Saturation and Medium-Window Diagnostics

The fully pooled robustness checks are retained as audit sanity checks rather than as primary evidence. They are deliberately conservative and mostly saturated: for DeepSeek top-10 adjacent-swap rank noise, ALFWorld ID and ToolBench all-split pooled recovery remain at 1.0 through $\epsilon = 0.5$, while ALFWorld OOD falls only at the most severe tested point, to 0.83 at $\epsilon = 0.5$. Under fixed step erasure, all reported ALFWorld and ToolBench pooled cells for both AgentMark-F and AsymMark-R recover at 1.0 through 90% erasure. Listing those cells as tables adds little beyond showing that the pooled audit window is overpowered for this diagnostic.

Table 2 instead reports a non-saturated path-consistency view from the fine ToolBench rerun. The metric is not full RLNC payload recovery: it is the probability that a short audit window keeps the decoded rank path unchanged under adjacent swaps. This is closer to the perturbation quantity used in Proposition 2 and exposes the expected decay with noise and window length.

TABLE 2. MEDIUM-WINDOW PATH CONSISTENCY UNDER ADJACENT-SWAP RANK NOISE. TOOLBENCH, DEEPSEEK, G1 INSTRUCTION SPLIT, 500 TRIALS PER CELL. VALUES ARE PATH-WINDOW SUCCESS RATES, NOT FULL PAYLOAD RECOVERY.

k	$\epsilon = 0.05$			$\epsilon = 0.25$			$\epsilon = 0.50$		
	4-step	8-step	12-step	4-step	8-step	12-step	4-step	8-step	12-step
3	.794	.656	.504	.498	.198	.090	.174	.034	.004
5	.728	.572	.406	.182	.042	.016	.064	.002	.000
10	.704	.502	.388	.246	.056	.016	.046	.000	.000

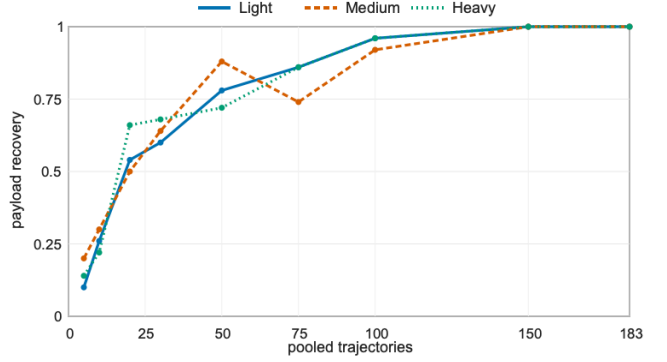


Figure 7. Semantic-rewrite pooled recovery for AsymMark-R on ToolBench, DeepSeek, top-10. Curves show sampled payload recovery as the audit window grows; the final point pools all 183 trajectories and matches the packets/conflicts endpoint in Table 3.

Appendix F. Semantic Rewrite Diagnostics

Rank perturbations isolate the channel mechanism, but a verifier may also face behavior serializations that are semantically close while textually changed. Table 3 reports the non-window diagnostics for the ToolBench semantic-rewrite rerun under the top-10 verifier view, while Fig. 7 plots the audit-window recovery curve. The rewrite strength changes the surface form substantially: token Jaccard falls from .532 under light rewrites to .271 under heavy rewrites. The exact rank path is therefore not preserved at the step level, and single trajectories almost never recover the 8-bit payload. Pooled recovery nevertheless rises as the verifier accumulates surviving packets, reaching 1.0 for all three strengths when the full 183-trajectory audit window is used.

Appendix G. Cross-Model Rank Calibration

We also run ToolBench cross-model calibration experiments in which the verifier reconstructs top- k ranks with a different model from the embedder. The same-model curve in Fig. 5 checks packet yield under matched conditions; the cross-model rerank checks whether the verifier model preserves rank paths well enough for pooled recovery. Figure 8 summarizes the live ToolBench operating point: DeepSeek-to-Gemini recovers with the shallower top-3 view as the audit window grows, while Gemini-to-DeepSeek needs the slightly deeper top-5 view.

The live L5 protocol also marks a domain boundary. On ALFWorld, cross-model top-10 overlap is moderate (0.596 for DeepSeek-to-Gemini and 0.633 for Gemini-to-DeepSeek), but top-1 agreement is only about 0.20, and pooled recovery succeeds only for the Gemini-to-DeepSeek ID pool in the pulled artifact. This reinforces the calibration rule rather than changing the main claim: rank-only decoding is useful when the verifier can reproduce rank paths for the deployed domain and model pair.

TABLE 3. SEMANTIC REWRITE DIAGNOSTICS FOR ASYMMARK-R ON TOOLBENCH, DEEPSEEK, TOP-10. SIMILARITY AND RANK METRICS ARE STEP-WEIGHTED OVER 359 REWRITTEN STEPS PER STRENGTH; STRICT DSR IS SINGLE-TRAJECTORY PAYLOAD RECOVERY OVER 183 TRAJECTORIES; PACKETS/CONFLICTS ARE MEASURED ON THE FULL POOLED AUDIT WINDOW.

Rewrite	ROUGE-L	Jaccard	Rank keep	Top-1 match	Top-10 overlap	Strict DSR	Packets/conflicts
Light	.519	.532	.306	.393	.593	.011	45/14
Medium	.304	.316	.309	.379	.593	.011	45/13
Heavy	.272	.271	.312	.409	.594	.011	45/14

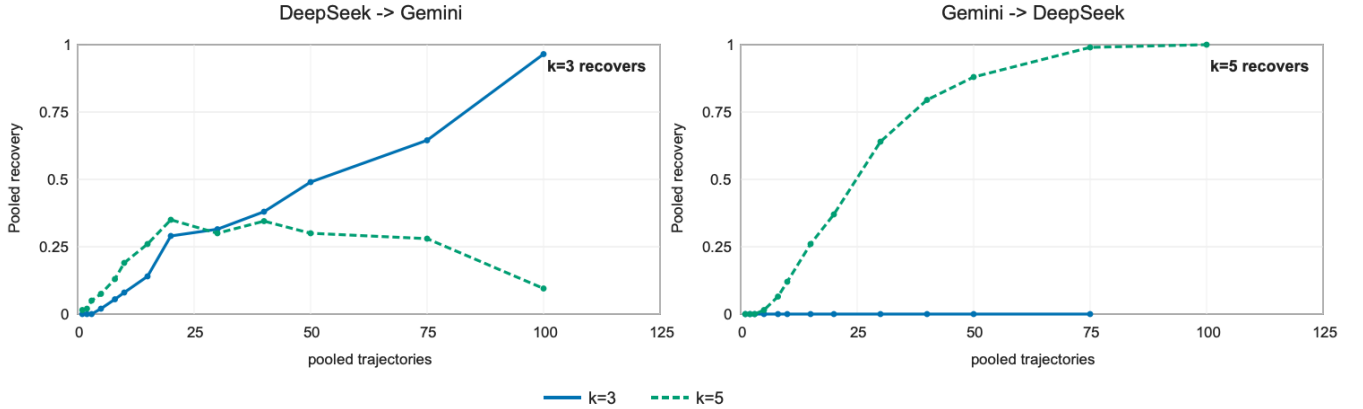


Figure 8. Live ToolBench L5 cross-model operating-point calibration. Curves show sampled pooled payload recovery for two rank depths under cross-model verification. DeepSeek-to-Gemini favors top-3, while Gemini-to-DeepSeek favors top-5, showing that verifier rank depth should be calibrated for the deployed model pair.

Appendix H. Evidence Confidence Diagnostics

The confidence-curve diagnostics in Fig. 9 illustrate the distinction between bit-level channel evidence and payload attribution. Bit-level evidence can become statistically strong before a full payload is recovered, but ownership attribution requires the same audit to satisfy claim binding, rank-quality gating, and pooled payload recovery.

Appendix I. Vanilla Baseline

The vanilla ALFWorld result is reported only as a task-performance sanity check for the explicit prompting and clean AgentMark pipeline used by the main comparisons. On the canonical 0510 ALFWorld artifact, the clean induced pipeline reaches 80.0% success with 16.23 steps, while the vanilla agent reaches 69.1% success with 16.54 steps. Table 4 expands this aggregate by model and split. This comparison is not a watermarking baseline; it shows that the explicit behavior-induction interface used to expose candidate actions does not degrade ALFWorld task performance before watermarking is applied.

TABLE 4. ALFWORLD VANILLA SANITY CHECK FOR EXPLICIT BEHAVIOR INDUCTION, EXPANDED BY MODEL AND SPLIT. SUCCESS AND STEPS ARE MEAN $\pm$ STANDARD DEVIATION OVER THE COMPLETED RUNS.

Method	Model	Split	Runs	Success (%)	Steps
Vanilla	Gemini	ID	3/3	60.95 $\pm$ 1.09	18.45 $\pm$ 0.07
Vanilla	Gemini	OOD	3/3	66.67 $\pm$ 1.55	17.67 $\pm$ 0.49
Vanilla	DeepSeek	ID	3/3	71.67 $\pm$ 1.65	15.59 $\pm$ 0.56
Vanilla	DeepSeek	OOD	3/3	77.11 $\pm$ 0.86	14.46 $\pm$ 0.39
Clean induced	Gemini	ID	3/3	72.62 $\pm$ 0.82	17.34 $\pm$ 0.08
Clean induced	Gemini	OOD	3/3	69.40 $\pm$ 0.00	18.76 $\pm$ 0.00
Clean induced	DeepSeek	ID	3/3	88.57 $\pm$ 0.00	14.64 $\pm$ 0.01
Clean induced	DeepSeek	OOD	3/3	89.55 $\pm$ 0.00	14.19 $\pm$ 0.03

Appendix J. Reproducibility Mapping

The reported tables are tied to postprocessed artifacts at the claim level:

- Utility and stealth claims are backed by task-success, step-count, and behavior-frequency summaries over the completed benchmark groups.
- Verifier-channel claims are backed by exact, rank-only, and selected-only decoder views over the same logged trajectory corpus.
- Mechanism claims are backed by a probability-bin counterfactual that holds rank order fixed while perturbing probability values.
- Top- k calibration claims are backed by offline rank-depth sweeps and live cross-model reranking studies.
- Robustness claims are backed by rank-noise and erasure studies using pooled audit-window recovery.

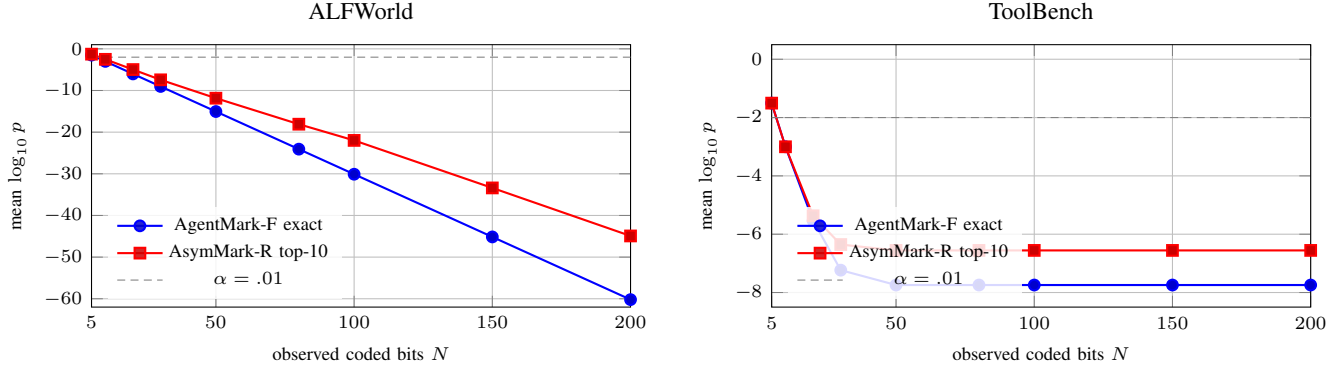


Figure 9. Bit-level evidence confidence under the binomial null. Curves aggregate completed model/split cells and report the mean one-sided $\log_{10} p$ value as the verifier observes more decoded coded bits. These curves size evidence windows; positive attribution still requires pre-registered payload recovery.

- Evidence claims are backed by random-claim false-positive trials and bit-level confidence diagnostics.

Machine-readable sidecars mirror these summaries for automated checks.

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